armfield

CEXC – Chemical Reactors Teaching Equipment – Service Unit

The Armfield CEXC family is a range of chemical reactors specifically developed for the teaching and demonstration of chemical reactor capabilities to chemical engineering students. Real chemical reactions take place within the reactors. Armfield has developed a number of representative reactions which are easy and safe for students to use in the laboratory environment.

Software allows remote control making distance learning easy, contact sales for more information.

PRODUCT CODE: **CEXC**

Categories: CE Series - Chemical Engineering, CEB-MKIII Transparent Batch Reactor, CEM-MKII Continuous Stirred Tank Reactor (CSTR), CET-MKII Tubular Reactor, CEXC – Chemical Reactors Teaching Equipment - Service Unit, CEY Laminar Flow Reactor, CEZ Plug Flow Reactor, Chemical, Educational

Tags: Batch Reactor , Benchtop , CE Series , CEXC , Chemical Engineering Basic Process Principles , Chemical Reactors Teaching Equipment , Colour Tracer , Compact , Computer Controlled , Continuous Stirred Tank Reactor , Data Logging , Glass Feed Vessels , K-Type Thermocouples , Laminar Flow Reactor , Peristaltic Pumps , PID , Plug Flow Reactor , Service Unit , Thermocouples , Transparent Reactors , Tubular Reactor

DESCRIPTION

The Armfield CEXC family is a range of chemical reactors specifically developed for the teaching and demonstration of chemical reactor capabilities to chemical engineering students. Real chemical reactions take place within the reactors.

Armfield has developed a number of representative reactions which are easy and safe for students to use in the laboratory environment.

The CEXC Chemical reactors teaching unit provides the services required to run the various reactor types. It includes; a hot water re-circulator used to control the temperature of the reactions, glass feed vessels for the reactants, two peristaltic pumps to pump the reagents to the reactors, computer software for data logging, sensors and instrumentation.

The CEXC is fully computer controlled, supplied with software to allow the user to; vary the feed pump speeds and flow rates, vary the heater power in the hot water, implement a PID control loop ensuring stable temperatures, switch on and off the hot water pump and to control the speed of the stirrers used on some of the reactors.

Instrumentation for temperature and conductivity measurements is also supplied and these values are displayed on the computer screen. Two 'K' type thermocouples are included; one for the hot water and one for the reactor contents.

Note: An input for a third user-supplied sensor is also provided for project work.

Full educational software is provided with the CEXC for all the Armfield chemical reactors.

Separate programs are provided for each reactor. Typical facilities are:

- All the temperatures and flow rates are displayed on a diagrammatic representation of the equipment
- The feed pump speeds are between 0 and 100%. The predicted flow rate for these speeds can be displayed and used for subsequent calculations. These values can be user calibrated for greater accuracy
- The stirrer speeds can be controlled in a similar manner
- The hot water temperature is set by entering a required temperature set point into a PID control function
- Data from the sensors is logged into a spreadsheet format
- Sophisticated graph plotting facilities are provided. Comparisons between data taken on different runs can be displayed
- Processing of measured values to obtain calculated values
- The data samples (measured and calculated) can be saved or exported directly in Microsoft Excel format
- Data from the sensors can be displayed independently from the data logging. This can be in bar graph format or a recent history graphical display

Presentation screens are available giving an overview of the software, the equipment, the procedure and the associated theory. This is backed up by a detailed 'Help' facility giving in-depth guidance and background information.

A dual-range conductivity sensor allows for a wide range of operation. Armfield has developed an algorithm for the saponification reaction (ethyl acetate and sodium hydroxide) linking the degree of conversion of the reactants to the electrical conductivity, thus allowing the progress of the reaction to be monitored using the software.

The service unit includes a mounting position for the reactor being used. It is possible to change reactors quickly and easily without the use of tools. All fittings on the CEXC and the reactors are of the quick-release type. The CEM-MKII, CET-MKII and CEB-MKIII reactors are completely contained on the CEXC base unit. The CEY and CEZ also include floor standing columns for positioning next to the CEXC base unit.

The CEXC provides a locating position for two standard 2.5 litre chemical storage bottles for the reagents within the plinth. This provides safety in use. The bottles can be quickly capped and removed as necessary for safe handling. Two 2.5 litre bottles are also provided with the equipment. Alternatively, for longer experiments, larger feed vessels could be located either on the floor or on the bench next to the equipment.

The user must have a PC with a USB port, running Windows 7 or above.

TECHNICAL SPECIFICATIONS

- A self-contained benchtop service unit designed to provide services for up to five different chemical reactors:- Continuous Stirred Tank Reactor – CEM-MKII- Tubular Reactor – CET-MKII- Transparent Batch Reactor – CEB-MKIII- Laminar Flow Reactor – CEY- Plug Flow Reactor – CEZ
- Fully computer controlled and supplied with educational software specific to each reactor type. Simple interfacing to the (user supplied) computer by a USB interface
- Two peristaltic feed pumps with individually variable flow rates 0-140 ml/min
- Provides PID temperature-controlled hot water in order to maintail reactor temperature
- Complete with two thermocouples, an input for a third (user) thermocouple and a dual-range conductivity sensor

- A comprehensive instruction manual is included which details installation and operating procedures

FEATURES & BENEFITS

Unique features

- Computer control and data logging as standard
- Remote control using the supplied software
- Compact benchtop equipment
- Real-time reaction monitoring instrumentation eliminating the inconvenience and inaccuracy of multiple iterations
- Safe and student friendly
- Transparent reactors enabling the student to see what is happening
- Colour tracer experiments possible for some reactor types
- Cost effective. Up to five reactors share the same service unit
- **Five different reactor types available:**

  1. Continuous stirred tank reactor – CEM-MKII
  2. Tubular reactor – CET-MKII
  3. Batch reactor – CEB-MKIII
  4. Laminar flow reactor – CEZ
  5. Plug flow reactor – CEY

Video

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Ordering Specification

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Optional Accessories

The CW-17 is a thermostatically controlled chilled water circulating unit which can be used for providing water at below ambient temperature. It is essential for use with the CEZ reactor and for the CEB-MkIII isothermal demonstration and can be used with the other reactors. The temperature of the water is controlled by an adjustable thermostat mounted on the CW-17. Alternatively, it can be used as an ice bank chiller to supply water slightly above freezing point.

CEXC Essential accessories table

Requirements

Electrical supply: Single phase

Software requires the user to have a PC running Windows 7 or above with a USB port.

At least 1 reactor – CEM-MKII, CET-MKII, CEB-MKIII, CEY, CEZ

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.4m³

Gross Weight: 45Kg

Dimensions

Length: 1.00m

Width: 0.50m

Height: 0.50m

Order Codes

CEXC-A: 220-240V / 1ph / 50Hz, 10 amp
CEXC-B: 120V / 1ph / 60Hz, 10 amp
CEXC-G: 220-240V / 1ph / 60Hz, 10 amp

CW-17-A: 220-240V / 1ph / 50Hz, 13 amp

CW-17-G:* 220-240V / 1ph / 60Hz, 13 amp

***CW-17-G** version has optional 3kVA transformer available to accommodate 120V / 1ph / 60Hz supply. If you require this option please specify at time of order.

YOU MAY ALSO LIKE…

CEM-MKII CONTINUOUS STIRRED TANK REACTOR (CSTR)

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CEZ LAMINAR FLOW REACTOR

Read more

CEB-MKIII TRANSPARENT BATCH REACTOR

Read more

CET-MKII TUBULAR REACTOR

Read more

CEY PLUG FLOW REACTOR

Read more

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